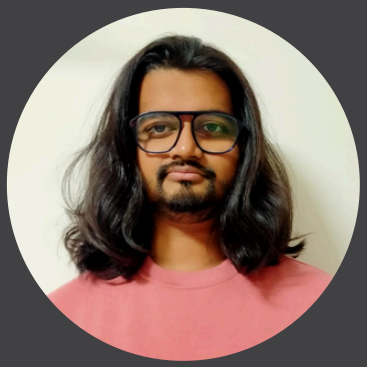


CONTACT

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SKILLS

- **Core AI / ML**
  - Machine Learning, Deep Learning
  - Computer Vision, NLP
  - CNNs, Transformers, Siamese Networks
  - Supervised, Unsupervised & Metric Learning
  - Data Analysis
- **Generative AI & LLMs**
  - LangChain, Vector Databases
  - RAG Pipelines, Agent-based Systems.
  - Prompt Engineering.
  - LLMs: GPT-style models.
- **Deep Learning Frameworks**
  - PyTorch, TensorFlow, Keras
  - Custom Model Architectures
  - Model Training, Evaluation & Optimization
- **ML-Ops, Deployment & Systems**
  - End-to-End ML Pipelines (Training → Deployment → Monitoring)
  - Model Deployment (Cloud & Edge)
  - Docker, Git
  - Performance Optimization & Inference Tuning
  - CI/CD for ML Pipelines
  - FastAPI, Sheldon
- **Cloud Platforms**
  - AWS, GCP, Microsoft Azure
- **Programming**
  - Python

Utkarsh Shrivastav

Senior AI/ML Engineer

PROFILE

Senior AI/ML Engineer with 6+ years of experience designing, building, and deploying production-grade **Computer Vision, Deep Learning, and Generative AI** systems across cloud and edge environments. Strong expertise in **CNNs, Transformers, LLM-based RAG pipelines, and end-to-end ML lifecycle (data → training → deployment → monitoring).**
Proven track record of delivering real-world AI solutions in **waste management, anti-counterfeiting, and intelligent automation**, including live deployments for global brands such as **Nestlé (Maggi Masala)** and **Harpic**. Experienced in model optimization, scalable system design, and cross-functional collaboration to translate complex business problems into robust AI products.

EXPERIENCE

May-2023 to Present
Ficha

SENIOR AI ENGINEER

- Designed, developed, and deployed production-grade machine learning and deep learning solutions using Python, PyTorch, and TensorFlow for computer vision and intelligent automation use cases.
- Built and maintained end-to-end ML pipelines covering data preprocessing, feature engineering, model training, evaluation, deployment, and production monitoring.
- Worked with cloud-based AI/ML deployment workflows, leveraging AWS and related cloud services where applicable to support scalable model serving and data processing.
- Applied MLOps practices to streamline model experimentation, versioning, deployment, performance monitoring, and lifecycle management.
- Developed LLM-based applications using OpenAI models, prompt engineering, retrieval-augmented generation (RAG), LangChain, and vector databases to enable contextual information retrieval and intelligent question answering.
- Processed, queried, and analyzed structured and unstructured datasets using Python and SQL to support data validation, exploratory analysis, and model development.
- Developed data processing and visualization workflows to analyze model performance, evaluate business metrics, and communicate actionable insights to stakeholders.
- Optimized model inference, runtime performance, and resource utilization for cloud and edge deployments, achieving a 3.5–5× inference speedup through pipeline and system-level improvements.
- Conducted model validation, error analysis, and performance benchmarking using appropriate evaluation metrics to improve prediction accuracy and production reliability.
- Collaborated with product, engineering, and operations teams to translate business requirements into scalable AI solutions and deliver projects from research and prototyping through production deployment.

Feb-2020 to April-2023
Ennoventure Technologies Pvt. Ltd, Bengaluru

ML ENGINEER

- Developed, trained, and optimized machine learning and deep learning models using Python, TensorFlow, and PyTorch for image classification, object detection, image similarity, and anti-counterfeiting applications.
- Built data preprocessing, image augmentation, feature extraction, and model evaluation pipelines to improve data quality, model robustness, and generalization.
- Designed and maintained reusable Python utilities to automate model training, testing, evaluation, and data processing workflows.
- Analyzed structured and unstructured data using Python and SQL, where applicable, to support data quality checks, reporting, and model performance analysis.

EDUCATION

2015-2019
BE - COMPUTER SCIENCE
CDGI, Indore, M.P, IN

CERTIFICATIONS

- **Introduction to TensorFlow for AI, ML, and DL** | Coursera | June 2021
- **Convolutional Neural Networks in TensorFlow** | Coursera | June 2021
- **Natural Language Processing in TensorFlow** | Coursera | June 2021
- **Sequences. Time Series and Prediction** | Coursera | June 2021
- **App Development (Windows platform using C#)** | Microsoft Training Program| September 2015
- **Web Development(using Python)| Microsoft Training Program**| April 2016

HOBBIES



- Applied MLOps and software engineering practices to improve experiment reproducibility, model testing, deployment reliability, and maintainability.
- Contributed to the development and deployment of production ML solutions supporting business applications for global FMCG brands, including Nestlé Maggi Masala and Harpic.
- Evaluated model performance using precision, recall, F1-score, and other relevant metrics, identifying failure cases and implementing improvements.
- Developed scalable computer vision pipelines for synthetic image generation, image filtering, and one-shot learning using Siamese Neural Networks.
- Collaborated with cross-functional teams to convert business requirements into production-ready ML solutions and support the complete project lifecycle

PROJECTS (FICHA)

COCON

The COCON is a connected, AI-powered container placed on the sorting bins within trash rooms in collective housing. It helps raise awareness of the correct sorting action by reducing sorting errors. Thanks to its artificial intelligence, it helps improve waste management efficiency and cleanliness of trash rooms and make sorting interactive and fun.

- Led research and development to evaluate and select optimal computer vision model architectures for object characterization and sorting error detection.
- Designed and trained task-specific deep learning models tailored to real-world waste classification and filtering scenarios.
- Deployed production-grade models across both cloud and edge environments, enabling scalable and low-latency inference in on-site installations.
- Optimized model performance and inference reliability through extensive testing, validation, and iterative refinement under real-world conditions.
- Collaborated with cross-functional teams (product, hardware, and engineering) to address system-level challenges and ensure seamless end-to-end integration.

CAMIA

CAMIA is an AI-powered vision system installed in dump trucks to characterize the sorting quality of each bin deposited and detect sorting errors. Equipped with an artificial intelligence system, it allows communities and collection operators to reduce sorting refusals on their territory.

- Led research and development to identify and benchmark optimal computer vision model architectures for waste characterization and sorting quality assessment.
- Designed and trained deep learning models to detect sorting errors and evaluate bin content under diverse environmental and operational conditions.
- Deployed production-ready models across cloud and edge environments, supporting low-latency inference directly on collection vehicles.
- Optimized model robustness and performance through extensive real-world testing, validation, and iterative improvement.
- Collaborated with cross-functional teams (hardware, data, and operations) to integrate AI models into end-to-end collection workflows and resolve system-level challenges.
- Addressed challenges related to motion, lighting variation, and constrained compute environments in mobile edge deployments.

LLM Chatbot

Developed an advanced chatbot using LangChain, Vector DB, and an agent framework to enhance user interaction and query resolution. LangChain was utilized for managing complex dialogue flows, Vector DB for efficient data retrieval, and the agent framework for dynamic response generation. This integration resulted in a chat bot capable of providing coherent, contextually relevant, and rapid responses to user inquiries.

- Utilized LangChain for handling complex dialogue and maintaining conversation context.
- Implemented Vector DB for fast and accurate data retrieval.
- Integrated an agent framework to enable adaptive and dynamic responses.

PROJECTS (ENNOVENTURE)

Invisible Cryptographic Signature (Maggi Masala & Harpic)

- Served as lead ML engineer for live anti-counterfeit systems leveraging deep learning-based image preprocessing, classification, and filtering.
- Integrated custom ML models into production systems used by global FMCG brands.

Image Classification & Filtering (CNN-based Systems)

- Designed and deployed CNN-based classification and filtering pipelines for real world image data across web and cloud platforms.
- Focused on model robustness, performance optimization, and deployment reliability.

Object Detection Systems (YOLO, SSD)

- Developed and deployed object detection models using YOLO and SSD architectures to identify regions of interest in real-world images.
- Optimized detection accuracy and inference performance for cloud-based applications.

Synthetic Image Generation & Augmentation

- Built computer vision pipelines for synthetic image generation by blending digital objects with real-world backgrounds.
- Applied advanced augmentation, enhancement, and realism techniques to improve training data quality and model generalization.

One-Shot Learning

One-Shot Learning & Similarity Modeling

- Led development of a one-shot learning system using Siamese Neural Networks to measure visual similarity between images.
- Designed models suitable for real-time inference and web-based deployment.